AEGISWAVE AI — MASTER SYSTEM SPECIFICATION

Sovereign, Multipurpose, Accurate, Reliable, Portable & Data-Secure AI Architecture

Target System: HP Victus Gaming Laptop (AMD Ryzen 5 5600H, NVIDIA GTX 1650 4GB VRAM, 8GB DDR4)
Workspace Path: D:\HiddenWave\Projects\My_Bot | **Document Version:** 1.0.0 Production Core

1. Core Philosophy & The Four Pillars

AegisWave AI is an autonomous personal AI engineered from first principles for complete data sovereignty, factual accuracy, reliability, and portability. It operates without streaming unredacted user prompts or confidential documents to cloud infrastructure, keeping full ownership and control in the user's hands.

Pillar	Core Mechanism	User Benefit
1. Data Secure	Real-time regex & entropy PII scrubbing; local SQLite storage.	Zero telemetry. Passwords, API keys, and client emails never leave unmasked.
2. Accurate	Empirical verification loop auditing numbers against tool observations.	Eliminates hallucinations. Math and facts are proven by code execution.
3. Reliable	Security sandbox jailing tool actions inside project boundaries.	Crash-resilient SQLite WAL checkpoints; code cannot harm the host OS.
4. Portable	Self-contained project folder with relative path resolution.	Entire bot and memory travel on a USB drive or laptop with zero install hassle.

2. Target Hardware Profile (HP Victus)

The architecture is explicitly tuned for your laptop's specific hardware constraints:

Component	Specification	Optimization Strategy
CPU	AMD Ryzen 5 5600H (6C/12T, 3.3-4.2 GHz)	Allocates up to 8 worker threads for data parsing and AST math.
GPU (Dedicated)	NVIDIA GeForce GTX 1650 (4 GB GDDR6)	Houses Q4_K_M 1B-3.8B models in VRAM for instant CUDA speed (~45 tok/s).
System RAM	8 GB DDR4 (Single Stick @ 3200 MHz)	Prevents RAM exhaustion by keeping heavy model weights on the GPU.
Storage	512 GB NVMe PCIe M.2 SSD	Rapid model loading (<2s) and high-concurrency SQLite WAL commits.

3. The 5-Stage Autonomous Execution Pipeline

Whenever a user submits an instruction, it traverses a deterministic 5-stage sovereign pipeline:

- **Stage 1: Security Shield:** Scans the prompt via regex/entropy patterns. Replaces API keys (sk_...), emails, and IP addresses with surrogate tokens (,). Logs audit action.
- **Stage 2: Reasoning Core:** Injects multi-turn conversational history from SQLite. Determines intent (conversational dialogue vs. tool action) and generates execution plan.
- **Stage 3: Sandboxed Tools:** Executes tools (read_file, calculate, system_telemetry, web_fetch). Path jailing strictly blocks attempts to escape D:\HiddenWave\Projects\My_Bot.
- **Stage 4: Accuracy Verifier:** Audits proposed output numbers and claims against raw tool observation logs. Flags or rejects hallucinations before user presentation.
- **Stage 5: Local Vault Commit:** Restores original secrets locally for user display. Commits message records and audit entries to SQLite WAL database.

4. Dual Engine Strategy & Capability Domains

AegisWave AI features a plug-and-play Model Router supporting both offline local execution and sovereign cloud supercharging:

Domain	Option A: Local Neural (Llama 3.2 3B)	Option B: Sovereign Cloud (Gemini/Groq)
Privacy & Air-Gap	100% Offline. Zero bytes leave machine.	Sanitized. Sensitive tokens masked by Shield first.
Text & Language	Strong daily writing, summarization, translation.	Massive context (1M+ tokens), handles 100+ languages.
Creativity	Good for copy, email drafting, brainstorming.	Master-level storytelling, tone nuance, scripts.
Data & Logic	Verified arithmetic via AST tool + Python debugging.	Deep multi-step algorithmic reasoning & complex math.
Vision & Sound	Requires local modules (Whisper/Moondream).	Native multimodal vision, audio, and handwriting OCR.
Automation	Path-jailed local filesystem & telemetry tools.	Identical local tools with higher-level planning.

5. User Interfaces & Operations

The system can be operated through three complementary interfaces:

1. **Universal Interactive Launcher:** python start_bot.py (Provides numbered menu for CLI, Web, Diagnostics, and Model Download).
2. **Rich Terminal CLI:** python bot/main.py (Interactive console with /telemetry, /audit, /vault, /clear commands).
3. **Mobile & Desktop Web Gateway:** python web/server.py (Flask REST API on Port 5000. Mobile phones access via local Wi-Fi or Tailscale PWA).
4. **Visual Architecture Guide:** python launch_gui.py (Interactive browser visualizer on Port 8765).

6. Complete Repository Structure

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D:\HiddenWave\Projects\My_Bot\ ■■■ start_bot.py # Universal Interactive Menu ■■■ download_model.py # Local Neural Model (Llama 3.2 3B) Downloader ■■■ launch_gui.py # Visual Architecture Server (Port 8765) ■■■ AEGISWAVE_MASTER_MANUAL.md # Full Markdown Specification ■■■ AegisWave_AI_Master_Manual.pdf # Official PDF Documentation ■■■ config/ ■ ■■■ hardware_profile.json # HP Victus GPU/CPU hardware tuning specs ■ ■■■ settings.py # Environment roots, sandbox rules, model defaults ■■■ bot/ ■ ■■■ main.py # Rich Interactive Terminal CLI Client ■ ■■■ core/ # agent.py (ReAct Loop), verifier.py (The Critic) ■ ■■■ security/ # shield.py (PII Scrubber), sandbox.py (Path Jail) ■ ■■■ memory/ # vault.py (SQLite WAL Storage) ■ ■■■ tools/ # files.py, system.py, math_tool.py, web.py ■ ■■■ models/ # router.py, gpt4all_provider.py, cloud_provider.py ■■■ web/server.py # Flask Mobile & Web Gateway (Port 5000) ■■■ gui_demo/index.html # Modern Visual Architecture Dashboard ■■■ data/ # vault.db (SQLite) & models/ (Local GGUF Weights)
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AegisWave AI — Document compiled and verified on local HP Victus hardware.